Exhaust – General Data

1. General

The exhaust system consists of two exhaust nozzles, one on each side of the engine exhaust case. The nozzles protrude through the engine cowlings, at a point between the upper and lower cowlings, then curve rearwards to direct the exhaust gases to the side of the engine and over the wing. To assist in the direction change of the exhaust gases, two curved vanes are fitted inside each exhaust nozzle near the outlet end. Each nozzle is secured to the engine exhaust case flange with nuts, bolts and washers. The cowlings have fairings around the nozzle openings.

2. Removal/Installations

A. Remove Exhaust Nozzle

1) Open lower engine cowling and remove upper engine cowling. (Refer to 71-10-00, Cowlings – General Data).

2) Remove nuts, bolts, and washers, securing nozzle to engine exhaust duct flange and remove nozzle.

B. Install Exhaust Nozzle

1) Position nozzle on engine exhaust duct flange and secure with nuts, bolts, and washers.

2) Install upper cowling and close lower engine cowling. (Refer to 71-10-00, Cowlings – General Data).

3. Approved Repairs

A. Weld Cracked Exhaust Nozzle

1) Refer to Para 2.A and remove nozzle.

2) Clean nozzle with a suitable cleaning solvent.

3) Stop-drill crack using a No. 36 drill and carefully deburr.

4) Clean weld area with a clean stainless steel brush or fine abrasive paper.

5) Carry out weld repair by T.I.G. welding method using Inconel 625 filler wire. Preheating is not required.

CAUTION DO NOT CARRY OUT LOCAL STRESS RELIEVING BY TORCH METHOD

6) Stand nozzle on flanged end and stress relieve at 1600 ± 25°F for one hour then air cool in still dry air.

Note Small repairs (1/4 inch maximum) do not require relief after welding.

7) Refer to Para 2.B and install nozzle.

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